

TRIGONOMETRY

Practice Sheet

SECTION – A

MCQ & Assertion–Reason

1 mark each

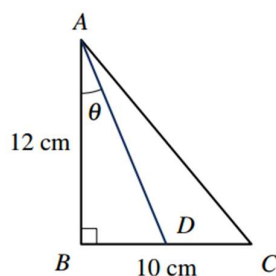
- If $4 \tan \theta = 5 \sin \theta$, where $0^\circ < \theta < 90^\circ$, then the value of $\sin^2 \theta - \cos^2 \theta$ is
 (a) $\frac{7}{25}$ (b) $-\frac{7}{25}$ (c) $\frac{1}{25}$ (d) $\frac{16}{25}$
- For an acute angle θ , which of the following is not possible?
 (a) $\sin \theta = \frac{3}{5}$ (b) $\sec \theta = \frac{1}{2}$ (c) $\tan \theta = 5$ (d) $\cot \theta = \sqrt{3}$
- If $\sin \theta = \frac{a}{b}$, where $0 < a < b$, then $\sec \theta + \tan \theta$ is equal to
 (a) $\sqrt{\frac{b+a}{b-a}}$ (b) $\sqrt{\frac{b-a}{b+a}}$ (c) $\frac{a+b}{b-a}$ (d) $\frac{b}{b-a}$
- Assertion (A) :** For every acute angle θ , $\sin \theta + \cos \theta \leq \sqrt{2}$.
Reason (R) : $(\sin \theta + \cos \theta)^2 = 1 + 2 \sin \theta \cos \theta$, and $2 \sin \theta \cos \theta \leq \sin^2 \theta + \cos^2 \theta$ for all θ .
 (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true but R is not the correct explanation of A
 (c) A is true but R is false.
 (d) A is false but R is true.

SECTION – B

Short Answer

2–3 marks each

- If $\tan \theta = 24$, find the value of $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$
- If $3 \sin^2 \theta + 7 \cos^2 \theta = 4$ and θ is acute, find θ . Hence find the value of $\tan \theta + \cot \theta$.
- Evaluate : $\frac{2 \sin^2 60^\circ + 3 \tan^2 30^\circ - \operatorname{cosec}^2 45^\circ}{\cot^2 30^\circ + \sec^2 60^\circ - \operatorname{cosec}^2 30^\circ}$
- If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$. Where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B.
- It is given that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ and $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. Using these, evaluate $\frac{3 \sin 10^\circ - 4 \sin^3 10^\circ}{4 \cos^3 10^\circ - 3 \cos 10^\circ}$.
- If $\sin^4 \theta + \cos^4 \theta = \frac{1}{2}$ and θ is acute, find the value of θ .
- In $\triangle ABC$, right-angled at B, $AB = 12$ cm and $BC = 10$ cm. D is the mid-point of BC and $\angle BAD = \theta$. Find:
 (i) $\sin \theta$ (ii) $\cos \theta$ (iii) $\tan \theta + \cot \theta$ (iv) $\cot^2 \theta - \operatorname{cosec}^2 \theta$



SECTION – C

Prove / Long Answer

3–5 marks each

12. Prove that $\frac{1}{1-\tan^2 A} + \frac{1}{1-\cot^2 A} = 1$, where $A \neq 45^\circ$.
13. Prove that $\frac{\sin \theta}{1-\cot \theta} + \frac{\cos \theta}{1-\tan \theta} = \sin \theta + \cos \theta$.
14. Prove that $\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} + \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = 2 \sec \theta$.
15. Prove that $(\sin A \cos B + \cos A \sin B)^2 + (\cos A \cos B - \sin A \sin B)^2 = 1$.
16. If $x = a \sec \theta + b \tan \theta$ and $y = a \tan \theta + b \sec \theta$, prove that $x^2 - y^2 = a^2 - b^2$.
17. If $\sec \theta + \tan \theta = l$, prove that $\sec \theta = \frac{l^2+1}{2l}$. Hence show that $\sin \theta = \frac{l^2-1}{l^2+1}$.
18. Prove that : $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \frac{1 + \cos \theta}{\sin \theta}$.
19. Prove that $(1 + \tan A + \sec A)(1 + \cot A - \operatorname{cosec} A) = 2$.
20. Prove that : $\frac{\tan^3 \theta}{1 + \tan^2 \theta} + \frac{\cot^3 \theta}{1 + \cot^2 \theta} = \sec \theta \operatorname{cosec} \theta - 2 \sin \theta \cos \theta$.
21. Prove that $\frac{\sin^4 \theta + \cos^4 \theta - 1}{\sin^6 \theta + \cos^6 \theta - 1} = \frac{2}{3}$.
22. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, prove that $x^2 + y^2 = 1$.
23. If $\operatorname{cosec} \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$, prove that $(m^2 n)^{2/3} + (m n^2)^{2/3} = 1$.
24. If $\operatorname{cosec} \theta = x + \frac{1}{4x}$, prove that $\operatorname{cosec} \theta + \cot \theta = 2x$ or $\frac{1}{2x}$.
25. If $\sin \theta + \sin^2 \theta = 1$, find the value of $\cos^{12} \theta + 3 \cos^{10} \theta + 3 \cos^8 \theta + \cos^6 \theta - 1$.

ANSWER KEY & HINTS

Section : A

1. (B) $-\frac{7}{25}$
2. (B) ($\sec \theta \geq 1$ always)
3. (A)
4. (a)

Section : B

5. $\frac{31}{17}$ (divide numerator and denominator by $\cos \theta$)
6. $\theta = 60^\circ$; $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$
7. $\frac{1}{6}$ (numerator = $\frac{1}{2}$, denominator = 3)
8. $A - B = 30^\circ, A + B = 60^\circ \Rightarrow A = 45^\circ, B = 15^\circ$
9. $\frac{\sin 30^\circ}{\cos 30^\circ} = \tan 30^\circ = \frac{1}{\sqrt{3}}$
10. $1 - 2\sin^2 \theta \cos^2 \theta = \frac{1}{2} \Rightarrow \sin \theta \cos \theta = \frac{1}{2} \Rightarrow (\sin \theta - \cos \theta)^2 = 0 \Rightarrow \theta = 45^\circ$
11. $BD = 5 \text{ cm}, AD = 13 \text{ cm}.$
 - (i) $\frac{5}{13}$
 - (ii) $\frac{12}{13}$
 - (iii) $\frac{5}{12} + \frac{12}{5} = \frac{169}{60}$
 - (iv) -1 (true for every angle, since $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$)

Section : C

12. Write in $\sin/\cos$: $\frac{\cos^2 A}{\cos^2 A - \sin^2 A} + \frac{\sin^2 A}{\sin^2 A - \cos^2 A} = \frac{\cos^2 A - \sin^2 A}{\cos^2 A - \sin^2 A}$.
13. $\text{LHS} = \frac{\sin^2 \theta}{\sin \theta - \cos \theta} - \frac{\cos^2 \theta}{\sin \theta - \cos \theta}$; use $a^2 - b^2 = (a - b)(a + b)$.
14. Multiply inside each root by $(1 - \sin \theta)$ or $(1 + \sin \theta)$: terms become $\frac{1 - \sin \theta}{\cos \theta}$ and $\frac{1 + \sin \theta}{\cos \theta}$.
15. Expand both squares; the cross terms cancel; group as $\sin^2 A(\cos^2 B + \sin^2 B) + \cos^2 A(\sin^2 B + \cos^2 B)$.
16. $x^2 - y^2 = a^2(\sec^2 \theta - \tan^2 \theta) - b^2(\sec^2 \theta - \tan^2 \theta)$; the $2ab \sec \theta \tan \theta$ terms cancel.
17. $\sec \theta - \tan \theta = \frac{1}{l}$. Add to get $\sec \theta$, subtract to get $\tan \theta$; then $\sin \theta = \frac{\tan \theta}{\sec \theta}$.
18. Replace the 1 in the numerator by $\operatorname{cosec}^2 \theta - \cot^2 \theta$ and factorise; numerator
 $= (\operatorname{cosec} \theta + \cot \theta)(1 + \cot \theta - \operatorname{cosec} \theta)$.
19. $\text{LHS} = \frac{(\cos A + \sin A + 1)}{\cos A \sin A} = \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A}$.

20. $\frac{\tan^3 \theta}{\sec^2 \theta} = \frac{\sin^3 \theta}{\cos \theta}, \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} = \frac{\cos^3 \theta}{\sin \theta}; \text{sum} = \frac{\sin^4 \theta + \cos^4 \theta}{\sin \theta \cos \theta} = \frac{1 - 2\sin^2 \theta \cos^2 \theta}{\sin \theta \cos \theta}.$
21. $\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$ and $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta.$
22. From $x \sin \theta = y \cos \theta$, put $x = \frac{y \cos \theta}{\sin \theta}$ in the first equation to get $y = \sin \theta$, hence $x = \cos \theta.$
23. $m = \frac{\cos^2 \theta}{\sin \theta}, n = \frac{\sin^2 \theta}{\cos \theta} \Rightarrow m^2 n = \cos^3 \theta, mn^2 = \sin^3 \theta.$
24. $\cot^2 \theta = \operatorname{cosec}^2 \theta - 1 = \left(x + \frac{1}{4x}\right)^2 - 1 = \left(x - \frac{1}{4x}\right)^2, \operatorname{socot} \theta = \pm \left(x - \frac{1}{4x}\right).$
25. $\cos^2 \theta = \sin \theta.$ Expression $= \cos^6 \theta (\cos^2 \theta + 1)^3 - 1 = (\cos^2 \theta (1 + \cos^2 \theta))^3 - 1 = (\sin \theta + \sin^2 \theta)^3 - 1 = 0.$

